

SPL-1 PROPOSAL · IIT, UNIVERSITY OF DHAKA

FaceMatrix

Facial expression recognition in C++ with PCA and eigenfaces

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To a computer, a face is a grid of numbers

Problem and motivation

Find the face in an image, and name its expression.

Neutral

Happy

Sad

Surprise

Disgust

Expressions carry meaning

Software can react to the user

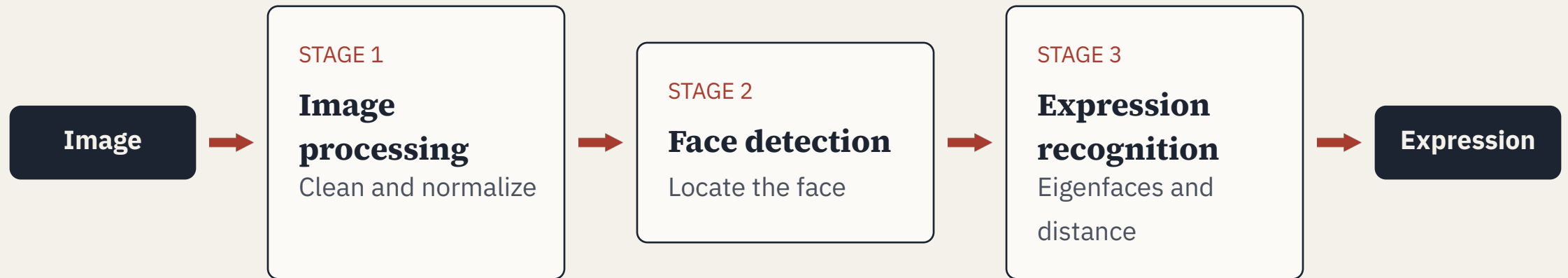
Pixels are not enough

Light, size and pose change them

Libraries hide the method

We want every step explained

Proposed solution



Method: Tawhid, Laskar and Ali, IJMLC 2012

Key features

From scratch

Standard C++ only, no external libraries

Own PCA solver

Eigenfaces computed by my own code

Five expressions

Neutral, happy, sad, surprise, disgust

Unknown rejection

Distance threshold instead of a forced guess

Inspectable

Mean face and eigenfaces saved as images

Fair evaluation

Real FER2013 data, confusion matrix

Target users

Students and teachers

See how face analysis works inside

Developers

A lightweight expression signal, no heavy dependencies

Researchers

A transparent classical baseline

Technology and tools

USING

- C++, standard library
- GNU Make / CMake
- FER2013 dataset (BMP)
- Unit tests per module

NOT USING

- OpenCV
- Eigen, LAPACK
- TensorFlow, PyTorch
- Pretrained models

FACEMATRIX

Thank you

Questions?

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